Data Quality and Harmonization

> Part of the Functional Series. The platform combines sources that were built for different purposes. This document describes, in non-technical terms, what the platform does to keep them trustworthy, what it will not do, and a one-page reference for all five sources.

1. The reference table

Source	Measures	Scope	Years	Updated	Licence	Main limitation
OWID / Global Carbon Project	CO_2 from fossil fuels and cement, by country; land-use CO_2 separately	CO_2 only	1750–2024	Annually (with the Global Carbon Budget); platform re-downloads monthly	CC BY 4.0 (OWID); cite the Global Carbon Project	Land-use CO_2 is modelled and uncertain; file mixes countries with regions
PRIMAP-hist v2.8	CO_2, CH_4, N_2O, F-gases by country, excluding land use	AR5 GWP-100, $MtCO_2e$	1750–2024	Annually (Sep/Oct)	CC BY-NC-SA 4.0	Composite reconstruction; no international transport; pre-1970 rebuilt
NOAA GML Mauna Loa	Atmospheric CO_2 concentration	Global, one station	1959–2025 annual; monthly to 2026	Monthly	Public domain	Single station
Law Dome ice core	Atmospheric CO_2 concentration, pre-1959	Antarctic air	1750–1958	Static	Cite the original papers	Smoothed; no per-year uncertainty
Berkeley Earth High-Res	Global surface temperature anomaly	Land + ocean	1850–2025	Monthly	CC BY-NC	Preliminary; baseline-dependent

A sixth source, EDGAR (the European Commission's emissions database), was evaluated and then **shelved**. It is retained only for internal cross-checks, not published, because part of its fuel-combustion CO_2 comes from the International Energy Agency under a no-derivatives licence that does not allow the platform to publish derived results.

2. The principles

One key, one unit, one scope. Every indicator in the harmonized layer is keyed by calendar year (and country, where relevant), has a single unit, and declares whether it is global or national. A figure never silently changes unit or scope between pages.

Gaps are gaps. Missing values stay missing. The platform does not interpolate across a gap or replace a missing gas with zero, because either would invent data.

Validate before publishing. Each source is checked before its output reaches the dashboard:

Check	Example from this platform
Completeness	A PRIMAP-hist year is published only if every gas covers at least 98% of the previous year's emissions
Plausibility	The total may not move more than ±15% from the previous year
Internal consistency	The gases must add up to the source's own total within 0.5%
Integrity	Downloads are verified against the publisher's checksums; a truncated file fails instead of publishing
Freshness	A stale or unusual source raises an alert but does not stop the run

When a check fails, the platform holds back the affected year or series. It does not repair it. That is why the latest PRIMAP-hist year shown is 2024 even though the file contains 2025.

Provenance is data. Every published series carries a record of where it came from: source URLs, retrieval time, the release version, a fingerprint of the raw file, coverage, units, scope, method, caveats and licence. The dashboard and API read this record; they do not recompute it.

Licences are enforced. Non-commercial terms, share-alike terms and citation requirements are recorded with each series, and derived data carries the notices it must. Using data under a licence that forbids a use is treated as a defect, which is how EDGAR came to be shelved.

3. Known discrepancies that are disclosed, not reconciled

Discrepancy	Why it exists
OWID/GCP and PRIMAP-hist CO_2 totals differ by a few percent	Different scope (international transport) and different energy statistics and emission factors
PRIMAP-hist is about 3% below EDGAR national totals by 2023, with large country differences (for example India about −14%, United States about +6%)	Different inventory construction and gap-filling
PRIMAP-hist country-reported and third-party scenarios differ by about 4.7% in 2023	Genuine uncertainty in the underlying data
The sum of countries differs from the World row in OWID	The World row includes international aviation and shipping; some summed entities are non-sovereign territories
Temperature "above pre-industrial" varies by a few hundredths of a degree between groups	Different baselines and different construction of the 1850–1900 reference

None of these is an error. They are the normal spread between competent datasets, and they set the realistic limit on how precisely any claim in this series can be made.

4. What this means for a reader

1. **Check the scope before comparing numbers.** Gases, land use, international transport, GWP set and baseline are the five things that explain most disagreements.
2. **Treat the latest year with care.** It is the one most likely to be revised, and for temperature it is flagged preliminary.
3. **Trust trends over single values.** The sources agree far better on direction and rate of change than on any one year's level.

See also

- Emissions Inventories
- Concentrations and Temperature
- Technical: Multi-Source Data Pipelines §3–5 — the validation, provenance and harmonization mechanics